

Osmania University

REGULAR EXAMINATION ? 2025

Reinforcement Learning ? Regular Examination, 2025

Subject Code: AI501 Semester: 6 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Markov Decision Processes)

1. Compare and contrast the key concepts involved in Markov Decision Processes. [12 Marks]
2. List the advantages and limitations of Markov Decision Processes. Explain any two in detail. [12 Marks]
3. List the advantages and limitations of Markov Decision Processes. Explain any two in detail. [12 Marks]

SECTION B (Unit 2: Dynamic Programming for RL)

4. Solve a problem related to Dynamic Programming for RL. Show all the steps clearly. [12 Marks]
5. Discuss the engineering applications of Dynamic Programming for RL. [12 Marks]
6. Solve a problem related to Dynamic Programming for RL. Show all the steps clearly. [8 Marks]

SECTION C (Unit 3: Temporal Difference Learning)

7. Define Temporal Difference Learning. Explain its significance in Reinforcement Learning with a suitable example. [12 Marks]
8. Write short notes on Temporal Difference Learning. [12 Marks]
9. Compare and contrast the key concepts involved in Temporal Difference Learning. [12 Marks]

SECTION D (Unit 4: Policy Gradient Methods)

10. List the advantages and limitations of Policy Gradient Methods. Explain any two in detail. [12 Marks]
11. Define Policy Gradient Methods. Explain its significance in Reinforcement Learning with a suitable example. [12 Marks]
12. Solve a problem related to Policy Gradient Methods. Show all the steps clearly. [12 Marks]

SECTION E (Unit 5: Deep Reinforcement Learning)

13. How is Deep Reinforcement Learning implemented? Give an algorithm / procedure and explain. [12 Marks]

14. List the advantages and limitations of Deep Reinforcement Learning. Explain any two in detail. [12 Marks]

15. Describe the working principle of Deep Reinforcement Learning with necessary equations. [12 Marks]

*** End of Question Paper ***